

LAB 3 – ARCHITECTURAL JUSTIFICATION

Academic Elective Bidding & Allocation System

Selected Architecture: Microservices Architecture

ARCHITECTURE SELECTION

We chose Microservices Architecture for the Academic Elective Bidding & Allocation System because its major functions, such as bidding, prerequisite validation, timetable checking, allocation, and academic records, can be developed as separate services. This makes the system easier to maintain, manage, and scale.

REASON 1 – SCALABILITY

The system must support 5,000 concurrent student sessions with response times below 3 seconds and process 5,000 bids in under 30 seconds. Microservices allows high-demand services such as Bidding Management and Allocation Solver to be scaled independently when required.

REASON 2 – FAULT ISOLATION AND MAINTAINABILITY

Prerequisite Validation and Timetable Conflict Checking can operate as separate services. A failure or update in one service is less likely to affect the entire system. Individual services can also be tested and deployed independently.

SECURITY ADVANTAGE

Microservices allows access control to be applied separately to each service. Sensitive student academic records and Registrar operations can be protected using authentication and authorization based on user roles.

PERFORMANCE BENEFIT

The Allocation Solver Service can be optimized and scaled independently during the allocation process. This helps the system process 5,000 bids within 30 seconds while keeping the student-facing services responsive during peak usage.